FITBUDDY

Complete Coding Documentation

Smart Fitness & Nutrition Assistant

README.md

```
# FitBuddy
Smart Fitness & Nutrition Assistant

Run:
1. Create and activate a virtual environment.
2. Install dependencies: pip install -r requirements.txt
3. Configure .env if an API key is required.
4. Run: uvicorn backend.main:app --reload
```

requirements.txt

```
fastapi
uvicorn
pydantic
python-dotenv
```

backend/main.py

```
from fastapi import FastAPI
from pydantic import BaseModel

app = FastAPI(title="FitBuddy API", version="1.0")

class WorkoutRequest(BaseModel):
    goal: str = "general fitness"
    level: str = "beginner"
    days: int = 7

class NutritionRequest(BaseModel):
    goal: str = "healthy eating"

@app.get("/")
def root():
    return {"message": "Welcome to FitBuddy"}

@app.post("/workout")
def workout(req: WorkoutRequest):
    plan = [
        {"day": 1, "focus": "Full body", "activities": ["Warm-up", "Squats", "Push-ups", "Cool-down"]},
        {"day": 2, "focus": "Cardio", "activities": ["Warm-up", "Brisk walk", "Stretching"]},
        {"day": 3, "focus": "Upper body", "activities": ["Warm-up", "Wall push-ups", "Light mobility"]},
        {"day": 4, "focus": "Recovery", "activities": ["Gentle stretching", "Rest"]},
        {"day": 5, "focus": "Lower body", "activities": ["Warm-up", "Bodyweight squats", "Glute bridge"]},
        {"day": 6, "focus": "Cardio & mobility", "activities": ["Easy cardio", "Stretching"]},
        {"day": 7, "focus": "Recovery", "activities": ["Light walk", "Stretching"]}
    ]
    return {"goal": req.goal, "level": req.level, "plan": plan[:req.days]}

@app.post("/nutrition")
def nutrition(req: NutritionRequest):
    tips = [
        "Include a variety of vegetables and fruits.",
        "Choose regular balanced meals.",
        "Drink water regularly.",
        "Include protein-rich foods appropriate to your diet."
    ]
    return {"goal": req.goal, "tips": tips}
```

frontend/index.html

```
<!DOCTYPE html>
<html>
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>FitBuddy</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <div class="container">
    <h1>FitBuddy</h1>
    <p>Smart Fitness & Nutrition Assistant</p>
    <label>Fitness Goal</label>
    <input id="goal" value="general fitness">
    <label>Level</label>
    <select id="level">
      <option>beginner</option>
      <option>intermediate</option>
      <option>advanced</option>
    </select>
    <button onclick="getWorkout()">Generate 7-Day Workout</button>
    <button onclick="getNutrition()">Get Nutrition Tips</button>
    <pre id="result">Your result will appear here.</pre>
  </div>
  <script src="script.js"></script>
</body>
</html>
```

frontend/style.css

```
body {
  font-family: Arial, sans-serif;
  margin: 0;
  padding: 30px;
  background: #f4f7f6;
}
.container {
  max-width: 650px;
  margin: auto;
  background: white;
  padding: 25px;
  border-radius: 12px;
}
input, select, button {
  width: 100%;
  padding: 10px;
  margin: 8px 0;
  box-sizing: border-box;
}
pre {
  white-space: pre-wrap;
  margin-top: 20px;
}
```

frontend/script.js

```
const API = "http://127.0.0.1:8000";

async function getWorkout() {
  const goal = document.getElementById("goal").value;
  const level = document.getElementById("level").value;

  const response = await fetch(`${API}/workout`, {
    method: "POST",
    headers: {"Content-Type": "application/json"},
    body: JSON.stringify({goal, level, days: 7})
  });

  const data = await response.json();
  document.getElementById("result").textContent =
    JSON.stringify(data, null, 2);
}

async function getNutrition() {
  const goal = document.getElementById("goal").value;

  const response = await fetch(`${API}/nutrition`, {
    method: "POST",
    headers: {"Content-Type": "application/json"},
    body: JSON.stringify({goal})
  });

  const data = await response.json();
  document.getElementById("result").textContent =
    JSON.stringify(data, null, 2);
}
```